Omni-Compass benchmark: Kubernetes native vs Kubernetes + Omni-Compass

Two identical kind clusters (1 control plane + 6 workers), same add-ons, workload (php-apache + HPA, target 50), load schedule and capture. **Native**: Omni-Compass not running. **Omni**: Omni-Compass drives the HPA target, the node pool (cordon/drain/uncordon), the power cap (in-place CPU limits on the app pods, enforced by the kernel), heat (harness law on live power), security (hold signal) and the rollout guard. Response times are real HTTP requests timed every 5 s. Power is a declared model (idle 100 W + 150 W x utilisation per worker in service; parked workers on standby at idle power), not a meter.

Gauges

Gauge	Native	Omni-Compass	Change
duration (min)	20.10	20.10	
worker nodes in service, mean	6.00	5.05	-15.8%
worker nodes in service, min	6.00	5.00	-16.7%
worker nodes in service, max	6.00	6.00	+0.0%
node-hours	2.01	1.69	-15.8%
power (W), mean	626	559	-10.8%
power (W), peak	652	621	-4.8%
energy (Wh)	210	187	-10.8%
energy, parked workers still on at idle power (Wh)	210	211	+0.5%
CPU used (cores), mean	0.816	0.888	+8.8%
CPU allocatable (cores), mean	24.00	20.20	-15.8%
utilisation (used / allocatable)	0.034	0.044	+29.3%
energy per core-hour (Wh)	768	629	-18.1%
node-hours per core-hour	7.36	5.69	-22.6%
pending pods, pod-minutes	0.250	0.267	+6.7%
pending pods, peak	1.00	1.00	+0.0%
HPA replicas, mean	9.00	8.86	-1.5%
HPA replicas, peak	10.00	10.00	+0.0%
HPA shortfall (desired > current), minutes	0.767	0.517	-32.6%
requests timed	1,125	1,145	+1.8%
time over the response line (% of samples)	0.000	0.000	0
response time (ms), mean	67.38	48.88	-27.5%
response time (ms), median	46.90	43.60	-7.0%
response time (ms), 95th percentile	153	72.18	-52.8%
response time (ms), 99th percentile	248	87.90	-64.6%
failed requests (%)	0.000	0.000	0

Significance (2-minute blocks: native 10, omni 10; bootstrap 95% interval)

Metric	Native	Omni-Compass	Change	95% CI of difference	Verdict
node_hours_per_core_hour	8.0339	6.5315	-18.7%	[-3.4909, +0.4728]	EVEN, no significant difference
kwh_per_core_hour	0.8352	0.7158	-14.3%	[-0.3080, +0.0736]	EVEN, no significant difference
utilisation	0.0337	0.0437	+29.7%	[-0.0015, +0.0226]	EVEN, no significant difference
pending_pod_minutes_per_hour	0.7258	0.7805	+7.5%	[-2.1774, +2.3415]	EVEN, no significant difference
hpa_shortfall_minutes_per_hour	2.2197	1.5000	-32.4%	[-5.1711, +3.0366]	EVEN, no significant difference

What Omni-Compass did

- decisions: 20 (one per minute)
- HPA target writes: 0
- node-pool resizes: 3
- scheduling-floor adds (pending pods): 0
- power-cap pod resizes (in place): 0
- rollout actions: 0
- power cap decided per minute: 1.0 1.0 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695 0.695
- heat (thermal) per minute: 0.358 0.391 0.416 0.436 0.455 0.472 0.485 0.499 0.51 0.52 0.528 0.533 0.537 0.541 0.545 0.549 0.552 0.554 0.556 0.557
- nodes decided per minute: 6 5 4 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3 3
- HPA target decided per minute (%): 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77 77

Reset

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restored target: 50
workers in service: 6 of 6
Omni records left after kill: none
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Minute by minute

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
0	6.00	6.00	616	617	5.00	5.00	0.000	0.000	0.551	0.561

min	nodes native	nodes omni	power W native	power W omni	replicas native	replicas omni	pending native	pending omni	CPU native	CPU omni
1	6.00	5.00	616	541	6.00	5.00	0.000	0.000	0.527	0.506
2	6.00	5.00	615	541	6.00	5.00	0.000	0.000	0.519	0.528
3	6.00	5.00	620	545	6.00	5.00	0.250	0.250	0.643	0.633
4	6.00	5.00	632	559	7.50	8.00	0.000	0.000	0.959	1.02
5	6.00	5.00	629	561	10.00	9.75	0.000	0.000	0.918	1.05
6	6.00	5.00	630	564	10.00	10.00	0.000	0.000	0.941	1.08
7	6.00	5.00	644	578	10.00	10.00	0.000	0.000	1.28	1.52
8	6.00	5.00	642	584	10.00	10.00	0.000	0.000	1.23	1.66
9	6.00	5.00	646	578	10.00	10.00	0.000	0.000	1.34	1.51
10	6.00	5.00	639	574	10.00	10.00	0.000	0.000	1.16	1.40
11	6.00	5.00	615	540	10.00	10.00	0.000	0.000	0.501	0.503
12	6.00	5.00	617	543	10.00	10.00	0.000	0.000	0.588	0.578
13	6.00	5.00	622	546	10.00	10.00	0.000	0.000	0.718	0.647
14	6.00	5.00	629	560	10.00	10.00	0.000	0.000	0.894	0.933
15	6.00	5.00	632	558	10.00	10.00	0.000	0.000	0.983	0.974
16	6.00	5.00	629	560	10.00	10.00	0.000	0.000	0.842	1.02
17	6.00	5.00	624	549	10.00	10.00	0.000	0.000	0.761	0.749
18	6.00	5.00	616	542	10.00	10.00	0.000	0.000	0.515	0.563
19	6.00	5.00	618	541	10.00	10.00	0.000	0.000	0.592	0.540

Limits

- kind nodes are containers sharing one CI machine; timings and CPU are noisier than real servers.
- The two arms ran on two different CI machines at the same time; machine-to-machine variation is part of the noise.
- Energy is modelled from utilisation and nodes in service with the declared constants above.